

04 - Flutter 基礎語法 01

一、Flutter 起手式

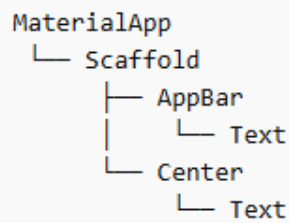
MaterialApp - App 的外層容器

Scaffold - 頁面骨架

AppBar - 內層上方標題列

AppBar - 內層下方內容區

Widget 組成關係：



```
import 'package:flutter/material.dart';

Run | Debug | Profile
void main() {
  var appTitle = Text('第一個Flutter App');

  var hiFlutter = Text('Hi, Flutter.', style: TextStyle(fontSize: 30));

  var appBody = Center(child: hiFlutter);

  var appBar = AppBar(title: appTitle);

  var app = MaterialApp(
    | home: Scaffold(appBar: appBar, body: appBody),
  ); // MaterialApp

  runApp(app);
}
```



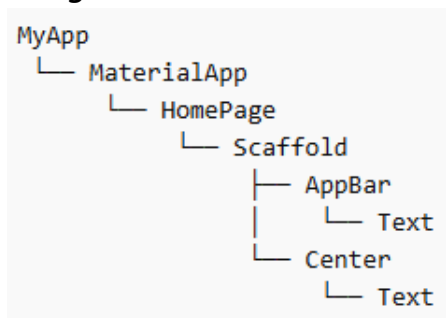
二、StatelessWidget 和 StatefulWidget

StatelessWidget - 不會變的畫面

StatefulWidget - 會改變的畫面

build() - 用來畫 UI，描述畫面長什麼樣子

Widget 組成關係：



```
import 'package:flutter/material.dart';
```

```
import 'package:flutter/material.dart';
```

Run | Debug | Profile

```
void main() {  
  runApp(const MyApp());  
}
```

```
class MyApp extends StatelessWidget {  
  const MyApp({super.key});
```

```
  @override
```

```
  Widget build(BuildContext context) {  
    return MaterialApp(  
      title: 'Hello Flutter',  
      home: const HomePage(),  
    ); // MaterialApp  
  }  
}
```

```
class HomePage extends StatelessWidget {  
  const HomePage({super.key});
```

```
  @override
```

```
  Widget build(BuildContext context) {  
    return Scaffold(  
      appBar: AppBar(  
        title: const Text('Hello Flutter!!'), // AppBar  
        body: const Center(  
          child: Text('Hello World!!', style: TextStyle(fontSize: 30)),  
        ), // Center  
    ); // Scaffold  
  }  
}
```



三、Text 、TextStyle 和 Color

Text 類別是建立字串物件，第一個參數必須是一個字串，後面再加上設定文字的屬性參數：

1.style：使用 TextStyle 類別物件設定

2.textAlign：

 TextAlign.left：靠左對齊

 TextAlign.right：靠右對齊

 TextAlign.center：靠中對齊

 TextAlign.justify：分散對齊

3.maxLines：限制最多顯示幾行，超過部分會被隱藏，若未設定文字長度預設會自動調整，字串換行可使用\n。

4.字串常用參數：

 fontSize：設定字體大小

 color：設定文字顏色

 decoration：文上加上底線

 fontWeight：文字粗細

backgroundColor : 文字背景顏色

```
class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key});

  @override
  Widget build(BuildContext context) {
    // TODO: implement build
    return Scaffold(
      backgroundColor:  Colors.yellow,
      appBar: AppBar(
        backgroundColor:  Colors.grey,
        title: const Text(
          "Change Text",
          style: TextStyle(backgroundColor:  Color(0xFF00FF00)),
        ),
      ),
      body: const Center(
        child: Text(
          "StatelessWidget",
          style: TextStyle(
            fontSize: 30,
            color:  Color(0xFFFF0000),
            decoration: TextDecoration.underline,
            fontWeight: FontWeight.bold, // FontWeight(900)
          ),
          textAlign: TextAlign.left,
          maxLines: 1,
        ),
      ),
    );
  }
}
```



四、Image

除了文字之外，影像也是 App 畫面經常出現的物件，可以使用 Image 類別建立影像，相關參數：

- 1.Image.asset - 載入本地端圖片
- 2.Image.network - 載入網路圖片
- 3.Image.file - 載入裝置端圖片

```

class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key});

  @override
  Widget build(BuildContext context) {
    // TODO: implement build
    return Scaffold(
      backgroundColor: Colors.yellow,
      appBar: AppBar(
        backgroundColor: Colors.grey,
        title: const Text(
          "Show Image",
          style: TextStyle(backgroundColor: Color(0xFF00FF00)),
        ),
      ),
      body: Center(
        child: Image.network(
          'https://miro.medium.com/1*cWklGIA01JspimzBensUKA.jpeg',
        ),
        child: Image.asset('assets/images/avatar.png'),

        child: Text(
          "StatelessWidget",
          style: TextStyle(
            fontSize: 30,
            color: Color(0xFFFF0000),
            decoration: TextDecoration.underline,
            fontWeight: FontWeight.bold, // FontWeight(900)
          ),
          textAlign: TextAlign.left,
          maxLines: 1,
        ),
      ),
    );
  }
}

```

```

class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key});

  @override
  Widget build(BuildContext context) {
    // TODO: implement build
    return Scaffold(
      backgroundColor: Colors.yellow,
      appBar: AppBar(
        backgroundColor: Colors.grey,
        title: const Text(
          "Show Image",
          style: TextStyle(backgroundColor: Color(0xFF00FF00)),
        ),
      ),
      body: Center(
        child: Stack(
          alignment: Alignment.topCenter,
          children: [
            Image.asset('assets/images/avatar.png'),
            const Text(
              "StatelessWidget",
              style: TextStyle(
                fontSize: 30,
                color: Color(0xFFFF0000),
                decoration: TextDecoration.underline,
                fontWeight: FontWeight.bold, // FontWeight(900)
              ),
            ),
          ],
        ),
      ),
    );
  }
}

```



五、使用 Center 和 Container 排列物件

Center：就是讓物件放在畫面中央位置，相關參數：

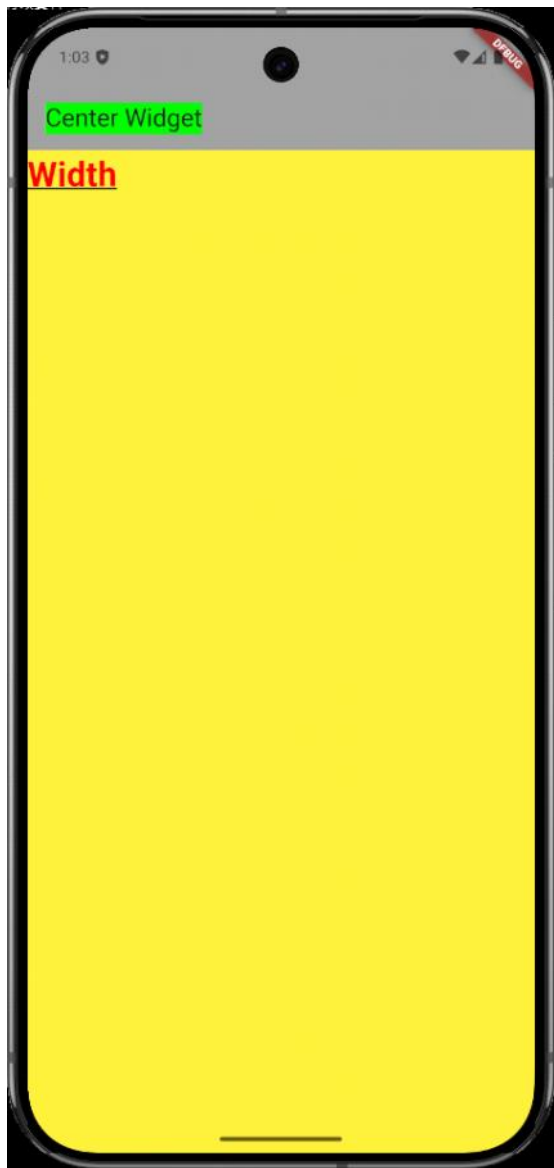
- 1.widthFactor - 指定排列範圍的寬度是物件寬度的幾倍
- 2.heightFactor - 指定排列範圍的高度是物件高度的幾倍

```

class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key});

  @override
  Widget build(BuildContext context) {
    // TODO: implement build
    return Scaffold(
      backgroundColor: Colors.yellow,
      appBar: AppBar(
        backgroundColor: Colors.grey,
        title: const Text(
          "Center Widget",
          style: TextStyle(backgroundColor: Color(0xFF00FF00))
        ),
      ),
      body: Center(
        heightFactor: 1,
        widthFactor: 1,
        child: Stack(
          alignment: Alignment.topCenter,
          children: [
            // Image.asset('assets/images/avatar.png'),
            const Text(
              "Width",
              style: TextStyle(
                fontSize: 30,
                color: Color(0xFFFF0000),
                decoration: TextDecoration.underline,
                fontWeight: FontWeight.bold, // FontWeight(900)
              ),
            ),
          ],
        ),
      ),
    );
  }
}

```



Container : 需要提供更多參數控制排列的方法，相關參數：

- 1.child - 指定要排列的物件
- 2.color - 背景色
- 3.alignment - 設定物件對齊方式
- 4.margin - 邊界
- 5.padding - 內距

```

class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key});

  @override
  Widget build(BuildContext context) {
    // TODO: implement build
    return Scaffold(
      backgroundColor: Colors.yellow,
      appBar: AppBar(
        backgroundColor: Colors.grey,
        title: const Text(
          "Hello Flutter!!",
          style: TextStyle(backgroundColor: Color(0xFF00FF00)),
        ),
      ),

      body: Center(
        child: Container(
          alignment: Alignment.topCenter,
          color: Colors.white,
          // margin: const EdgeInsets.fromLTRB(10.0, 20.0, 30.0, 40.0),
          margin: const EdgeInsets.all(20.0),
          // padding: const EdgeInsets.fromLTRB(10.0, 20.0, 30.0, 40.0),
          padding: const EdgeInsets.all(50.0),

          child: const Text(
            "Flutter App\nContainer Widget",
            style: TextStyle(
              fontSize: 30,
              color: Color(0xFFFF0000),
              decoration: TextDecoration.underline,
              fontWeight: FontWeight.bold, // FontWeight(900)
            ),
            textAlign: TextAlign.center,
          ),
        ),
      ),
    );
  }
}

```



六、使用 Row 、 Column 和 Stack 排列物件

Row：水平對齊，相關參數：

1. `mainAxisAlignment` - 主軸（水平）對齊方式
2. `crossAxisAlignment` - 副軸（垂直）對齊方式

Column：垂直對齊，相關參數：

1. `mainAxisAlignment` - 主軸（垂直）對齊方式
2. `crossAxisAlignment` - 副軸（水平）對齊方式

Stack：將全部物件疊在一起

```

class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key});

  get child => null;

  @override
  Widget build(BuildContext context) {
    // TODO: implement build
    return Scaffold(
      backgroundColor: Colors.yellow,
      appBar: AppBar(
        backgroundColor: Colors.grey,
        title: const Text(
          "Container Widget",
          style: TextStyle(backgroundColor: Color(0xFF00FF00)),
        ),
      ),
      body: Container(
        margin: const EdgeInsets.all(20.0),
        color: Colors.white,
        height: 300.0,
        width: 300.0,

        // transform: Matrix4.rotationZ(0.5), //3.1415926*角度/180
        transform: Matrix4.translationValues(50, 50, 0),

        child: Row(
          mainAxisAlignment: MainAxisAlignment
            .center, //start, end, center, spaceBetween, spaceAround,
            //spaceEvenly
          crossAxisAlignment:
            CrossAxisAlignment.stretch, //start, end, center, stretch
          children: [
            Expanded(
              flex: 3,
              child: Container(
                // margin: const EdgeInsets.fromLTRB(15, 10, 15, 10),
                child: const Text(
                  "Flutter1",
                  style: TextStyle(

```

